Uniformly accelerated rectilinear motion

Goal

After the video the student can find the velocity and displacement of a uniformly accelerated body from initial velocity, acceleration and time, and check the answer by the area under the v(t) graph.

The idea in brief

If speed never changes, motion is uniform: v equals s over t.
Here speed grows — but grows by the same amount each second.
Every second adds three metres per second — that is what an acceleration of three means.
Such motion is called uniformly accelerated.
Acceleration measures how fast velocity grows: a equals the change in velocity divided by time.
Its unit is the metre per second squared.
We point the x axis along the motion: everything forward takes the plus sign.
Velocity at any moment is the starting velocity plus the gain.
The gain is acceleration times time: v equals v naught plus a t.
Check the units: metres per second squared times seconds gives metres per second.
Displacement is how far the body shifted from the start.
While direction stays the same, it equals the distance travelled.
The slope of the line is the acceleration: the steeper, the larger the a.
Remember: the slope of the v–t graph equals the acceleration.
The area under the velocity graph numerically equals the displacement.

Quantities and units

v_0	initial velocity	2 m/s
a	acceleration	3 m/s²
t	time	4 s
v	final velocity	14 m/s
s	displacement	32 m

Formulas and conditions

Definition of acceleration:

$$a = \Delta v / \Delta t$$

Velocity:

$$v = v_0 + at$$

Displacement:

$$s = v_0 t + at^2/2$$

Displacement via average velocity:

$$s = (v_0 + v)t/2$$

- Motion is rectilinear, the x axis points along the motion.
- Acceleration is constant in magnitude and direction.

- Velocity keeps its sign — no reversal, displacement equals distance.
- All quantities in SI: m, s, m/s, m/s².

Figures

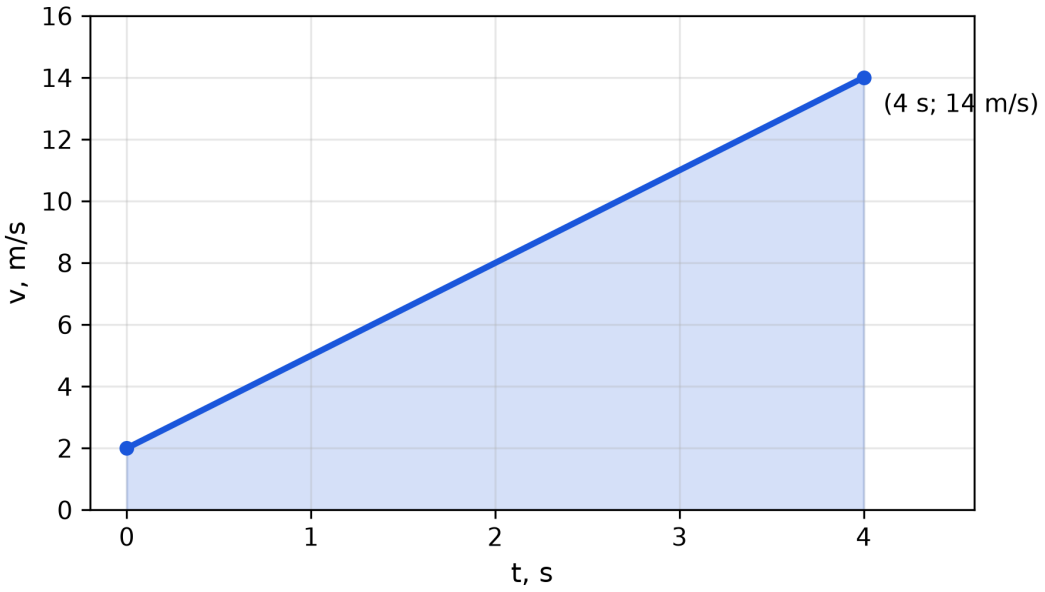


Figure 1: Graph of $v(t) = 2 + 3t$: the slope is the acceleration $a = 3$; the shaded area is the displacement $s = 32$.

Worked example

Given: $v_0 = 2$ m/s, $a = 3$ m/s², $t = 4$ s. Find: v, s.

1. $v = v_0 + at = 2 + 3 \cdot 4 = 14$ m/s.
2. $s = v_0 t + at^2/2 = 2 \cdot 4 + 3 \cdot 16/2 = 8 + 24 = 32$ m.
3. $s = (v_0 + v)t/2 = (2 + 14) \cdot 4/2 = 32$ m.

Units check: m/s² · s = m/s — consistent.

Traps

- Trap 1: $14 \cdot 4 = 56$ — wrong: the speed was growing, take the average.
- Trap 2: slope is acceleration, area is displacement.

Practice tasks

1. An object starts at 2 m/s and accelerates at 3 m/s². What is its speed after 4 seconds? **A)** 14 m/s **B)** 12 m/s **C)** 9 m/s **D)** 6 m/s

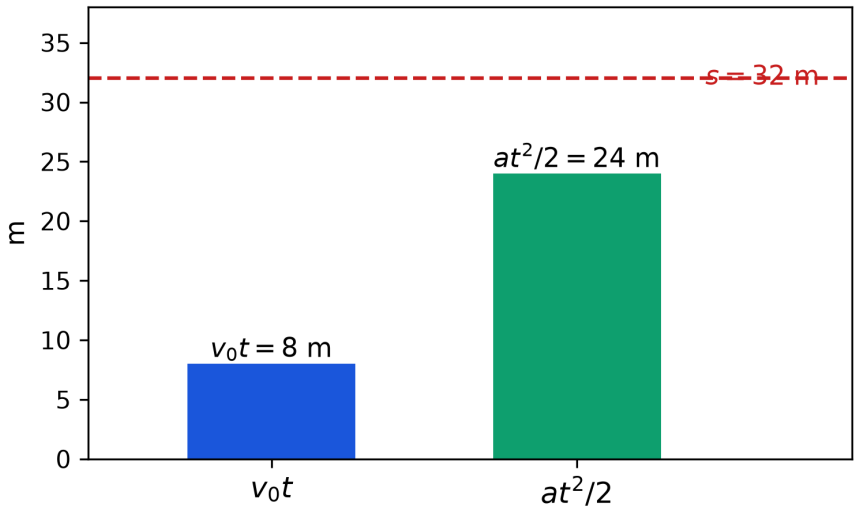


Figure 2: Displacement in two terms: $v_0t = 8$ plus $at^2/2 = 24$; total $s = 32$ m.

2. An object moves with initial speed 2 m/s and acceleration 3 m/s² without changing direction. Find its displacement in 4 s. **A)** 32 m **B)** 56 m **C)** 24 m **D)** 20 m

3. Which motion is called uniformly accelerated? **A)** Velocity changes by equal amounts in equal times **B)** Velocity stays constant **C)** Acceleration changes over time **D)** Distance is the same in equal times

4. What does the slope of the line on a velocity–time graph show? **A)** Acceleration **B)** Displacement **C)** Velocity **D)** Time

5. On a v(t) graph a straight line runs from (0 s; 2 m/s) to (4 s; 14 m/s). Find the displacement. **A)** 32 m **B)** 48 m **C)** 56 m **D)** 24 m

6. A car starts from rest with acceleration 2 m/s². How far does it go in 5 s?

Answers

1. **A** — Acceleration adds 3 m/s of speed each second. So $v = v_0 + at = 2 + 3 \cdot 4 = 14$ m/s.
2. **A** — $s = v_0 t + at^2/2 = 2 \cdot 4 + 3 \cdot 16/2 = 8 + 24 = 32$ m. 56 m comes from plugging the final speed into the uniform-motion formula.
3. **A** — In uniformly accelerated motion velocity gains equal increments in equal times while acceleration stays constant (Zakirova 9, §3). D describes uniform motion.
4. **A** — The slope of v(t) equals acceleration: $a = \Delta v/\Delta t$. Displacement is the area under the graph, not the slope (Zakirova 10, §3).
5. **A** — Trapezoid area: $(2 + 14)/2 \cdot 4 = 32$ m. 24 m is the triangle alone, without the rectangle.
6. **25 m** — $s = 0 \cdot 5 + 2 \cdot 25/2 = 25$ m.

Sources

Definitions, formulas and graph rules are cross-checked against the textbooks; numbers are PDF page numbers. Guide tasks are original.

- Физика Закирова 9.pdf — definitions and formula table: PDF с. 19–29.
- Физика Закирова 10 ЖМ.pdf — graphs and the area rule (§3): PDF с. 15–18, 20.
- FBS (1).pdf — formula and term cross-check (Sailaubaev): PDF с. 6–7.
- Testcenter -3.pdf — ENT graph-area task format (task 1, p. 2; key B, p. 37): PDF с. 2, 37.
- NCT physics sample (ru): https://new.testcenter.kz/wp-content/uploads/2024/12/obr_var_test_02.pdf
- NCT physics sample (en): https://new.testcenter.kz/wp-content/uploads/2025/06/PHYSICS.pdf

Status: physics and keys independently checked (Stage G); translations and terms need teacher review.